

BIUST Smart Maintenance System — Final Comprehensive Specification

This document provides a complete, bulletproof specification of the BIUST Smart Maintenance (BSM) System. It fuses all agreed features into a coherent design, ensuring clarity for anyone unfamiliar with the system. It covers public reporting, private operations, finance, inventory, notifications, and block/area management.

1. System Overview

- **Purpose:** A Computerised Maintenance and Management System (CMMS) for BIUST University.
 - **Architecture:** Two sides:
 - **Public Reporting Side:** Students and staff submit issues and track progress.
 - **Private Operations Side:** Campus assistants, operators, technicians, and coordinators manage workflows.
 - **Database:** PostgreSQL (pgAdmin, migrating to Prisma).
 - **Frontend Stack:** React + Vite, Tailwind CSS, Zustand, React Query, React Router, Jest.
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2. Authentication & Access

- **Public Side:**
 - Login via block → room → digital key.
 - Keys reset each semester.
 - On login, system auto-loads student details from CSV import.
 - **Private Side:**
 - JWT login with RBAC.
 - Finance module requires extra PIN unlock.
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3. Roles and Permissions

Role	Submit Report	View Tickets	Assign Tasks	Manage Inventory	View Analytics	Finance Access
Student	Yes	Own only	No	No	No	No
Staff	Yes	Own only	No	No	No	No
Campus Assistant	Yes (campus-wide)	All open	No	No	No	No
Operator	No	All	Yes	Read	Limited	No
Technician	No	Assigned	No	Update usage	No	No
Coordinator	Yes	All	Yes	Full	Full	Yes (PIN)

4. Block & Area Management

- **Initial Population:**
 - Coordinators upload CSV/Excel at semester start.
 - File includes student details (name, ID, omang, level, block, room, digital key).
 - System auto-creates blocks and rooms.
 - **Dynamic Management:**
 - Coordinators use "Manage Blocks & Areas" tool.
 - Add new blocks mid-semester.
 - Define rooms and assign students manually or via smaller CSV.
 - Blocks instantly appear in login dropdowns and notification selectors.
 - **Editing & Archiving:**
 - Rename, merge, or archive blocks.
 - Archived blocks remain in history but are hidden from active use.
 - Audit logs track all changes.
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5. Public Reporting Side

- **Login Flow:**
 - Student selects block and room.
 - Enters digital key.
 - System verifies and loads pre-imported details.
 - Personalized greeting: "Welcome John Doe."
 - **Resident Dashboard:**
 - Submit new report (title, description, category).
 - No priority selection — operators assign priority.
 - View tickets with detailed progress timeline.
 - Notifications feed (room-specific and scoped campus alerts).
 - Technician scheduling: resident selects availability → technician confirms.
 - **Duplicate Prevention:**
 - Prevent duplicate pending issues per room.
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6. Ticket Progress Timeline

- **Stages:**
 1. Report Submitted
 2. Operator Review
 3. Sourcing Funds
 4. Sourcing Materials
 5. Technician Assigned
 6. Scheduled Visit
 7. Work In Progress
 8. Completed
 9. Closed

- **UI Behavior:**
 - Horizontal timeline with labeled steps.
 - Completed = green checkmark.
 - Current = blue highlight.
 - Pending = grey.
 - Notes per stage.
 - **Backend Model:**
 - progress_stage enum.
 - progress_history with timestamps.
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7. Private Operations Side — IDE Workspace

- **Layout:**
 - Top toolbar: Search, alerts, reports, settings.
 - Left sidebar: Tickets, Job Cards, Inventory, Projects, Assets, Suppliers, Analytics, Finance.
 - Right sidebar: Filters, notes, quick actions.
 - Bottom panel: Logs, system messages.

Operator View

- Assign technicians.
- Set priority.
- Merge duplicates.
- Schedule preventive maintenance.
- View technician workload.

Technician View

- View assigned jobs.
- Complete job cards (digital PPCF form).

- Upload photos.
- Confirm visit times.
- Update inventory usage.

Campus Assistant View

- Submit campus-wide reports.
- View all open tickets.
- Escalate issues.
- Send alerts to operators/coordinators.

Coordinator View

- Full access to all modules.
- Room import (evacuate rooms, upload CSV/Excel).
- Manage blocks/areas dynamically.
- Preventive maintenance calendar.
- Targeted notifications (select blocks/areas or "Select All").
- Finance module (PIN-locked).

8. Inventory Management

- Table: item name, category, quantity, min threshold, status.
- Auto-update when job cards consume materials.
- Filtering/search.
- Low stock alerts.

9. Finance Dashboard (Coordinator Only, PIN-Locked)

- **Unlock Flow:**
 - Click Finance → enter PIN → unlock for session.
- **Budget Summary:**

- Total, allocated, remaining, burn rate.
 - **Spending Trends:**
 - Line graph with filters.
 - **Project Costs:**
 - Planned vs. actual.
 - **Forecasting:**
 - Overbudget risk graph.
 - **Procurement Tracker:**
 - Supplier orders linked to job cards.
 - **Budget Management:**
 - Create/edit budgets.
 - Transfer funds.
 - Log expenses.
 - **Project Dashboard:**
 - Milestones, progress, budget.
 - **Supplier Directory:**
 - Approved vendors, contracts.
 - **Compliance & Reporting:**
 - Audit logs.
 - Data export (CSV/PDF).
 - Safety inspections.
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10. Notifications

- **Coordinator Control:**
 - Choose target blocks/areas.
 - "Select All" for campus-wide alerts.

- **Resident Filtering:**
 - Only see alerts relevant to their block.
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11. Security & Privacy

- RBAC enforcement.
 - Finance PIN unlock.
 - Audit logs.
 - CAPTCHA and rate limiting.
 - Sensitive data minimized.
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12. Testing & Usability

- Unit tests for forms, dashboards, timeline.
 - Integration tests for API calls.
 - Coverage $\geq 70\%$ by Week 3.
 - Heuristic evaluation by Week 8.
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13. Folder Structure

frontend/

src/

components/

pages/

hooks/

store/

services/

utils/

public/

14. University-Wide Features

- Asset registry: track equipment and buildings.
- Supplier management: approved vendors, contracts.
- Compliance tracking: safety checks, audits.
- Resident feedback: satisfaction ratings post-completion.
- Disaster response mode: special workflow for emergencies.
- Multi-semester history: archive past tickets/projects.
- Data export: CSV/PDF reports for leadership.
- Mobile access: responsive design for technicians.

This specification is complete, detailed, and implementation-ready. It defines a full CMMS platform for BIUST University, covering maintenance, finance, analytics, communication, and block/area management across all roles.

Example Wireframes;

BIUST Smart Maintenance System

Search

Alerts

Cards

Reports

Settings

Help

Generate Report

Tickets

Job Cards

Inventory

Projects

Assets

Suppliers

Analytics

Finance

Tickets Dashboard

AssignedPriorityStatus

Ticket ID	Tatus	Priority	Status	Busy

Technician Load

Available

Busy

Merge Duplicates

Schedule PMS

Schedule PMs

Activity Log

Console

User logged in

Ticket updated

BIUST Smart Maintenance System

Search

Alerts

Cards

Reports

Settings

Help

Generate Report

My Jobs

Job Cards

Inventory

Assets

Suppliers

Analytics

Finance

Job Card #1234

Location : Block A, Room 102

Tasks

Materials Used

Add Photos

Upload Photo

Upload Photo

Schedule Visit

308712344

1	2	3	4
1	1	1	18
1	1	1	18
1	1	1	18
1	1	1	18
1	1	1	18
1	1	1	18
1	1	1	18
1	1	1	18

Notes

Add Note

Geotagged

Activity Log

Console

Log Instruct. Work on Job 1234

BIUST Smart Maintenance System

Search

Alerts

Cards

Reports

Settings

Help

Generate Report

Tickets

Job Cards

Inventory

Projects

Assets

Suppliers

Analytics

Finance

Campus Issues

Report New Issue

Escalate Issue

IssueLocationStatus

Broadcast Message

Send Alert

Quick Actions

Activity Log

Console

User logged in

Message sent to operators

Coordinator View with Finance Unlock

Search

Alerts

Cards

Reports

Settings

Help

Generate Report

Tickets

Job Cards

Inventory

Projects

Assets

Suppliers

Analytics

Finance

Finance Dashboard

Budget Overview

Spending Trends

Project Costs

Enter PIN to Unlock

PIN

Unlock

Project Milestones

Draw Alerts

Supplies

Expenses

Procurement

Project Milestones

View Projects

Compliance Checks

Activity Log

Console

BIUST Smart Maintenance System

Search

Alerts

Cards

Reports

Settings

Help

Tickets

Job Cards

Inventory

Projects

Assets

Suppliers

Analytics

Finance (Unlocked)

Budget Summary

Total Budget: 500,000

Allocated: 350,000

Remaining: 150,000

Monthly Burn Rate: 70,000

Spending Trends

All ProjectsJanJanJul

Procurement Tracker

Order	Supplier	Status	Linked To
PO-001	Vendor A	Delivered	#1234
PO-004	Vendor B	Approved	#1256
PO-003	Vendor B	Approved	#1256
PO-006	Vendor C	Requested	#1278

Transfer Funds

Move Budget

Log Transfer

Expense Log

Invoice #456\$2,500

Receipt #789\$800

Compliance Checklist

Safety Inspections

Audit Due: 01/15/22

Project Costs

Planned Costs

Actual Costs

Current Projects

Overbudget Risk

Current Projects

Project Name	Budget	Progress
Dorm Renovation	200,000	
IT Upgrade	100,000	
Roof Repairs	50,000	

Audit Log | Crassle

Budget adjusted by Admin

PO-001 marked as Delivered

Export CSV

Export PDF